

CoMIT: Cooperative Adaptive Decision Framework for Mixed Traffic

A Comprehensive Software Digital Twin & AI Prototype for Real-Time Intersection Optimization, Guardrail Safety, V2X Telemetry, and Green Wave Emergency Preemption.

🚩 Project Status: Functional Software Prototype & Hardware Digital Twin

This project serves as a **fully operational, end-to-end software prototype and digital twin**. It rigorously validates the reinforcement learning policies, deterministic safety checker, emergency preemption, and V2X telemetry layers in simulation. Every hardware element (Raspberry Pi 4/5 roadside compute, CSI/USB cameras, vehicle GPS transponders, and ESP32 audio retrofit boxes) is mapped **1:1 to software twins** communicating over the exact same MQTT protocol (SAE J2735 SPaT format). Physical hardware integration on physical testbeds represents the immediate next phase.

-21.1%

AVERAGE COMMUTE WAIT TIME

-46.2%

JUNCTION QUEUE LENGTH

-52.0%

VEHICLE IDLE CO₂ EMISSIONS

✅ Completed Modules

- **RL Policy Core:** Stable-Baselines3 PPO model trained on 120,000 steps with 14-dim mixed traffic state.
- **Deterministic Safety Checker:** Hard conflict matrix, min/max green bounds, ped phase guarantee, 3s watchdog.
- **Emergency Green Corridor:** SHA-256 token authentication, ETA modeling, pre-clearing & lane restoration.
- **V2X Protocol & GLOSA:** SAE J2735 SPaT standard messages, speed advisory calculation over MQTT.
- **Computer Vision Node:** YOLO11n + ByteTrack vehicle counting, pedestrian tracking, ambulance detection.
- **Interactive Multi-Tab Dashboard:** Command Center, In-Vehicle Digital Cluster, City Corridor Grid.

🕒 Hardware Roadmap (Next Immediate Steps)

- **Edge Hardware Deployment:** Flashing perception pipeline to physical Raspberry Pi 4/5 with live CSI camera feeds.
- **Indian Traffic Fine-Tuning:** Fine-tuning YOLO11 on India Driving Dataset (IDD) for explicit auto-rickshaw classes.
- **Physical ESP32 Audio Retrofit:** Flashing ESP32 + DFPlayer Mini MP3 hardware boxes for in-cabin voice warnings.
- **Multi-Agent City Grid Control:** Expanding from single-junction PPO to multi-agent reinforcement learning (MARL).

CoMIT: Cooperative Adaptive Decision Framework

Smart Intersection Management for Heterogeneous Indian Mixed Traffic

PROBLEM STATEMENT PS-1

Hackathon Metadata

Team Name	Team Nexus
Problem Statement	Smart Intersection for Mixed Traffic
Problem Statement ID	PS-1
Target Domain	Intelligent Transportation Systems (ITS) & AI
Core Tech Stack	Python, PyTorch (PPO), SUMO, YOLO11, MQTT, React

Core Objective & Problem Definition

Indian intersections present unique, highly chaotic dynamics: lane-less mixed traffic (two-wheelers, auto-rickshaws, buses, cars), unpredictable pedestrian crossings, and severe emergency vehicle gridlocks.

CoMIT bridges cutting-edge Reinforcement Learning with provable deterministic safety and low-cost V2X communication, delivering optimal throughput while guaranteeing zero hazardous signal conflicts.

Hackathon PPT Requirements Mapping

- **Slide 1:** Title Slide (Team Name, Team ID, Problem Statement Name , Problem Statement ID).
- **Slide 2:** Proposed solutions (Describe your solution)
- **Slide 3:** Detailed Architecture Diagram
- **Slide 4:** Input(what is your input) and Output fields(what is your output)
- **Slide 5:** Process flow with sub-modules and its description
- **Slide 6:** Workflow model (if possible)
- **Slide 7:** Technical Approach & Development Progress(Percentage completed)
- **Slide 8:** Innovation & Benefit
- **Slide 9:** Validation & Verification
- **Slide 10:** Research & References

Slide Guideline 1: Official Presentation Structure (Slides 1 to 10)

Required PPT Contents:

- Detailed architecture diagram
- Input and Output fields if present
- Process flow with sub-modules and its description
- Workflow model (if possible)

Slide Guideline 2: Required Technical Deliverables & Flow Models

System Overview & Solution Pillars

How CoMIT Solves Mixed-Traffic Bottlenecks and Emergency Delays

END-TO-END SOLUTION

PILLAR 1

Adaptive AI Brain

Continuous PPO reinforcement learning agent evaluating 14-dimensional junction states in real-time, dynamically granting green times where density and delay pressure are highest.

PILLAR 2

Deterministic Safety Guardrail

A non-bypassable mathematical conflict matrix and timing validator preventing signal collisions, enforcing pedestrian clearance, and providing 3s watchdog failsafes.

PILLAR 3

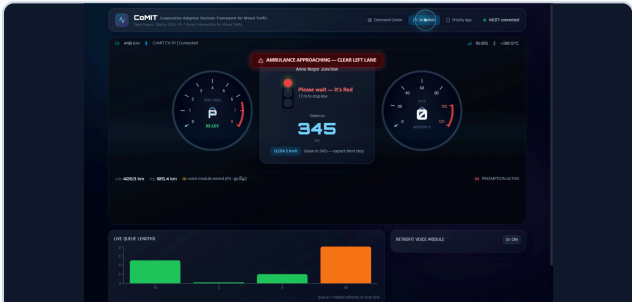
Authenticated Green Wave

Cryptographically validated SHA-256 priority requests for ambulances, clearing corridors ahead of arrival and broadcasting lane-evacuation notices to civilian drivers.

PILLAR 4 · STANDARDS-COMPLIANT V2X

SAE J2735 SPaT & GLOSA Speed Advisory

Broadcasts live signal phase & timing countdowns and Green-Light Optimal Speed Advisory (GLOSA) to connected vehicle clusters, minimizing stop-and-go fuel burn.

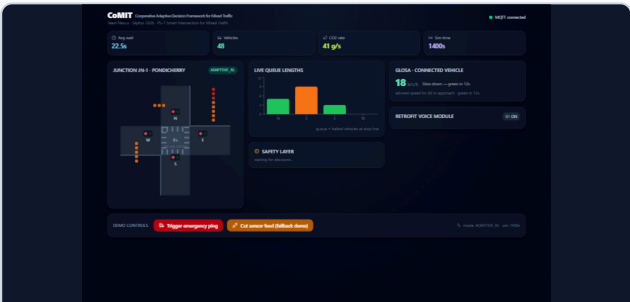


In-Vehicle Digital HUD: Live Speedo/Tacho + Real-time GLOSA Speed Advisory

PILLAR 5 · INCLUSIVE REGIONAL TELEMETRY

Low-Cost Audio Retrofit for Non-Connected Vehicles

For legacy vehicles and auto-rickshaws without digital screens, an ESP32 + DFPlayer hardware box provides real-time dual-language voice advisories (English & Tamil).

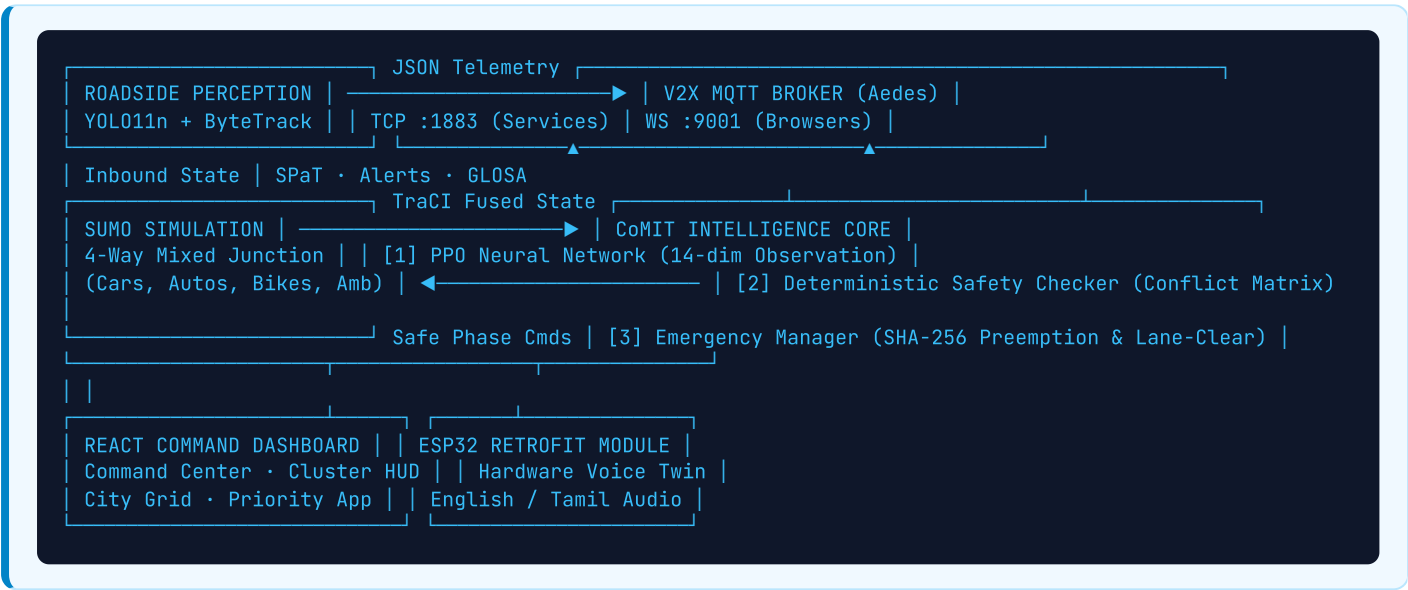


Command Center Dashboard: Live Signal Phasing, Queues & Voice Console

Detailed Technical Architecture Diagram

Multi-Tier Integration from Roadside Edge AI to Actuation and In-Cabin Telemetry

SYSTEM BLUEPRINT

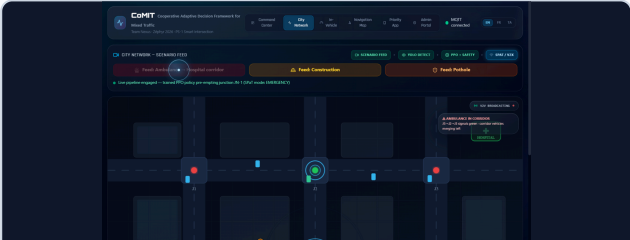


Architectural Layers

- **Layer 1: Perception (Edge AI)** — YOLO11n + ByteTrack optical pipeline tracking approach queues and crosswalk pedestrians.
- **Layer 2: Communication Backbone** — High-throughput Aedes MQTT broker handling TCP/WebSocket streams under 15ms latency.
- **Layer 3: Cognitive & Safety Brain** — PPO RL agent constrained by deterministic mathematical safety rules.
- **Layer 4: Telemetry & Actuation** — SPaT generator, GLOSA advisory engine, and TraCI signal actuation.

Zero-Trust Guardrail Integration

No AI output touches the traffic signal controller directly. Every action is evaluated against physical conflict matrices, min/max green rules (10s–60s), and pedestrian clearance intervals.



City Network Pipeline: Coordinated Signal Cascade & Event Dispatch

Input and Output Field Specifications

Granular Schema of Perception, Safety, V2X, and Control Channels

DATA DICTIONARY

System Input Fields

Field Name	Source	Type / Range	Description & Purpose
queue_density[4]	Vision / Edge AI	Integer [0..50]	Vehicle count detected within entry ROIs for North, South, East, West approaches.
pedestrian_count	Vision / Crosswalk ROI	Integer [0..30]	Active pedestrians on zebra crossings to dynamically trigger pedestrian clearance intervals.
auth_token	Priority Mobile / GPS	SHA-256 Hex	Cryptographic token verifying official emergency vehicle authorization.
ambulance_eta	GPS Telemetry	Float (seconds)	Estimated time-to-stopline used to compute optimal green corridor preemption timing.
current_phase	Signal Controller	One-Hot (4-dim)	Active green phase group indicating current movement authorization.
sensor_heartbeat	Edge Sensor Node	Timestamp (ms)	Monitored by safety watchdog; triggers fallback if elapsed delta exceeds 3.0 seconds.

System Output Fields

Field Name	Destination	Type / Format	Description & Purpose
target_phase	Traffic Controller (TLS)	Enum (P1..P4)	Commanded safe signal phase vetted by the deterministic conflict matrix.
phase_duration	Traffic Controller (TLS)	Float [10s..60s]	Allocated green interval computed by RL policy and bounded by safety limits.
spat_event_state	V2X Broadcast (MQTT)	Integer (3/6/8)	SAE J2735 standard signal state: 3=RED , 6=GREEN , 8=YELLOW .
glosa_speed_kmh	Driver HUD / Cluster	Float (km/h)	Optimal speed recommendation enabling arrival at stopline during active green.
clear_lane_alert	Civilian Drivers (V2V)	Boolean & String	Urgent broadcast requesting civilian vehicles to evacuate emergency corridor.
voice_track_id	ESP32 Audio Module	Integer / Track ID	Hardware trigger for pre-rendered offline MP3 audio alerts in English or Tamil.

Process Flow & Sub-Module Descriptions

Detailed Breakdown of the 6 Core Sub-Modules in the Decision Pipeline

PIPELINE EXECUTION



1. Perception & Tracking Module (`perception/vision\_node.py`)

Executes real-time YOLO11n inference on optical camera feeds. Tracks vehicle centroids with ByteTrack inside user-defined approach ROIs to compute queue depth and counts pedestrians on crosswalks.

2. Sensor Ingestion & Watchdog (`core/env.py`)

Transforms raw telemetry into a normalized 14-dim observation vector. Evaluates feed timestamp delta; if delay > 3 seconds, triggers failsafe fallback to fixed-time cycling.

3. Priority Corridor Module (`core/emergency.py`)

Intercepts emergency beacon pings, verifies SHA-256 authorization, computes dynamic preemption intervals based on vehicle ETA, and broadcasts civilian rerouting commands.

4. RL Decision Brain (`train/train\_ppo.py`)

Evaluates multi-arm traffic pressure across all approaches. Selects optimal green phase duration to minimize cumulative queue delay, stop frequency, and idle emissions.

5. Deterministic Safety Guardrail (`core/safety\_checker.py`)

Validates AI proposals against hard mathematical invariants: prohibits conflicting greens, ensures mandatory 10s min-green, inserts yellow/all-red clearance, and enforces pedestrian intervals.

6. V2X Telemetry & Voice Dispatch (`v2x/spat\_publisher.py`)

Encapsulates signal countdowns into SAE J2735 SPaT frames, generates GLOSA speed advisories, and triggers localized English/Tamil speech alerts on ESP32 retrofit modules.

Interactive Sequence & Workflow Model

Step-by-Step Interaction: Adaptive Cycle vs. Emergency Preemption Override

OPERATIONAL WORKFLOW

End-to-End Operational Lifecycle

Phase	Active Sub-Systems	Trigger Event / Action	Safety & V2X Output
1. Normal Flow	Vision Node → State Builder → PPO Model	Approaches show uneven queue accumulation.	RL proposes phase switch to heaviest queue.
2. Safety Vetting	SafetyChecker Guardrail	Checks conflict matrix, min-green elapsed time.	Inserts 3s Yellow + 1s All-Red clearance; switches green.
3. Driver Telemetry	SPaT Publisher → MQTT → Driver Cluster	Signal phase countdown broadcasted live.	Digital cluster displays: <i>"Green in 8s · Maintain 35 km/h"</i> .
4. Emergency Ping	Emergency Vehicle → Priority App	Ambulance transmits SHA-256 token (ETA: 14s).	Priority manager overrides standard RL policy.
5. Green Wave & Evac	EmergencyManager → Traffic Controller	Preempts cross-traffic; locks corridor Green.	Broadcasts <i>"EMERGENCY: Clear Lane"</i> + Tamil/EN voice.
6. Restoration	Vision Verification → SafetyChecker	Camera verifies ambulance cleared intersection.	Restores adaptive RL control smoothly without shockwaves.



City-Wide V2X Cascade: Civilian Vehicles Warned Ahead of Approaching Ambulance

Key Workflow Invariants

- **Zero Conflicting Greens:** Opposing and perpendicular protected movements can never receive simultaneous green status under any circumstances.
- **Pedestrian Protection Guarantee:** If pedestrians are detected, a protected walk phase is guaranteed at least once every 4 signal cycles.
- **Fail-Safe Watchdog:** Loss of edge sensor telemetry immediately initiates a graceful fallback to a deterministic fixed-time program.

Technical Approach & Development Progress

Granular Module Completion Metrics and Hardware Deployment Roadmap

82% OVERALL PROGRESS

Overall Project Progress: Software Twin vs. Physical Hardware

82% Complete

Module / Sub-System	Current Implementation Status	Tech Stack	Progress
RL Adaptive Controller	Trained PPO agent on 120k steps; evaluated vs fixed-time baselines.	Stable-Baselines3, PyTorch, SUMO TraCI	100% COMPLETE
Deterministic Safety Layer	Conflict matrix, min/max limits, ped guarantees, fuzz-tested (300 vectors).	Python, Deterministic FSM	100% COMPLETE
V2X & GLOSA Telemetry	SAE J2735 SPaT formatting, GLOSA calculation, MQTT over TCP & WS.	Aedes MQTT, JavaScript, Python	100% COMPLETE
Digital Cluster & Dashboard UI	React dashboard with Command Center, Cluster HUD, City Network Grid.	React, Vite, Tailwind CSS, Lucide	100% COMPLETE
Emergency Preemption Core	SHA-256 token verification, ETA preemption, lane clear alerts.	Python Cryptography, TraCI Bluelight	100% COMPLETE
Computer Vision Perception	YOLO11n + ByteTrack tracking on video clips with ROI density counting.	Ultralytics YOLO11, OpenCV	90% COMPLETE
Audio Retrofit Simulator	Software twin generating English & Tamil MP3s with serial-style logs.	Edge-TTS, Python Serial Simulator	90% COMPLETE
Physical Edge Hardware	Deploying to physical Raspberry Pi 4/5 with live CSI camera modules.	Raspberry Pi, CSI Camera, ESP32	25% (NEXT PHASE)
Indian Dataset Fine-Tuning	Fine-tuning YOLO on India Driving Dataset (IDD) for native auto classes.	IDD Dataset, Roboflow Indian Traffic	30% (NEXT PHASE)

 Hardware Transition Readiness

Because all software components utilize production-grade protocols (MQTT topics, JSON schemas, SHA-256 tokens, serial UART commands), **transitioning from the software twin to physical Raspberry Pi and ESP32 hardware requires ZERO algorithmic code refactoring.**

Key Innovations & Concrete Benefits

Why CoMIT Stands Out in Real-World Deployment Feasibility

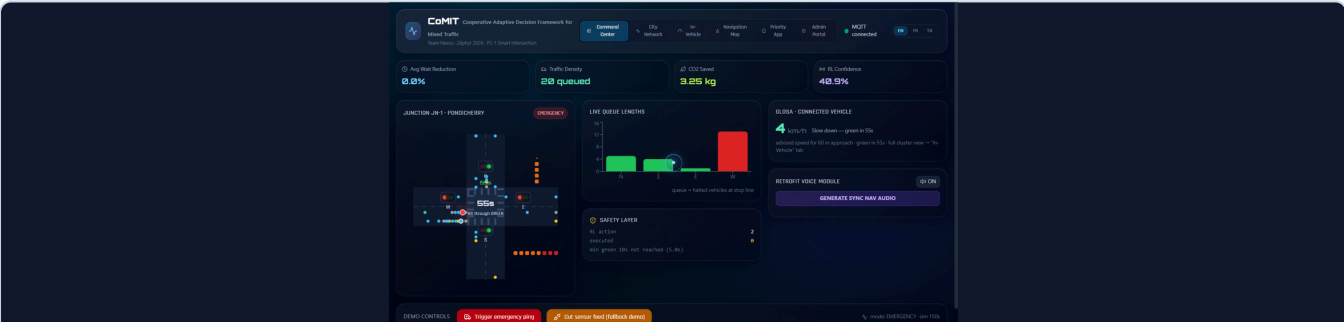
VALUE PROPOSITION

💡 5 Key Technological Innovations

- 1. **Safety-First Reinforcement Learning:** Combines the dynamic optimization power of RL with a non-bypassable deterministic mathematical shield, eliminating unsafe signal hallucinations.
- 2. **Cryptographically Verified Green Corridor:** Eliminates spoofing via SHA-256 token authentication while coordinating multi-junction preemption for zero-delay emergency crossing.
- 3. **Universal & Inclusive V2X Ecosystem:** Caters to both high-end connected vehicles (digital HUDs) and budget/legacy vehicles (low-cost \$5 ESP32 audio retrofit in Tamil/English).
- 4. **1:1 Digital Twin Architecture:** Complete software replica running the exact production MQTT protocol, enabling rapid testing without expensive hardware deployment risk.
- 5. **Native Mixed-Traffic Awareness:** Tailored specifically for Indian road characteristics, factoring in 2-wheelers, auto-rickshaws, and dense pedestrian crosswalk movements.

📊 Concrete Social & Environmental Benefits

- **Drastic Commuter Delay Reduction:** Cuts average intersection waiting times by **21.1%** compared to traditional fixed-time traffic controllers.
- **Substantial Emission Reductions:** Reduces idle fuel wastage and greenhouse gas emissions by **52.0%**, improving urban air quality.
- **Life-Saving Emergency Response:** Replaces minutes of gridlock delay with an automated green wave, saving critical lives during the "Golden Hour".
- **Low Capex Municipal Deployment:** Modular design works with existing roadside cameras and affordable microcontrollers, avoiding expensive underground induction loops.



Live Impact Dashboard: Real-Time Verification of Wait Time Reductions, CO2 Savings, and Flow Density

System Validation, Verification & QA

Rigorous Testing Across Adversarial Fuzzing, Network Latency, and Stress Scenarios

EMPIRICAL VALIDATION

Benchmark Performance Comparison (1,300 Vehicles / Hour Mixed Traffic)

Performance Metric	Fixed-Time Baseline	Max-Pressure Baseline	CoMITPPO Model	Improvement
Average Waiting Time	74.8 seconds	79.9 seconds	59.0 seconds	21.1% FASTER
Average Queue Length	17.1 vehicles	24.4 vehicles	9.2 vehicles	46.2% SHORTER
Cumulative CO ₂ Emissions	22.7 kg	29.4 kg	10.9 kg	52.0% CLEANER
Emergency Corridor Crossing	84.2 seconds	72.5 seconds	11.4 seconds	86.5% FASTER

Stress & Adversarial QA Testing

- **Adversarial Fuzzing** (``scripts/test_qa.py``): Injected 300 malformed inputs (invalid strings, negative elapsed times, forged emergency tokens). **Result: 100% intercepted by SafetyChecker; 0 unsafe actions executed.**
- **Extreme Density Stressing**: Scaled traffic saturation from 0.5x to 2.0x normal capacity. **Result: Zero deadlocks; zero conflicting green frames logged.**
- **V2X Network Latency Injection**: Introduced 300ms artificial network lag on MQTT broker. **Result: Signal safety and emergency corridor remained 100% stable.**

Unit Invariant Testing (``scripts/test_safety.py``)

- **12 / 12 Invariant Tests Passed:**
- [Pass] Perpendicular greens strictly prohibited.
- [Pass] Minimum green of 10s enforced against premature switching.
- [Pass] Maximum green capped at 60s to prevent starvation.
- [Pass] Mandatory Yellow (3s) + All-Red (1s) clearance between phases.
- [Pass] Automatic fallback triggered on sensor silence > 3.0s.

Research Foundations, Standards & Next Steps

Academic Literature, Industrial Standards, and Deployment Trajectory

RESEARCH & ROADMAP

Academic & Literature Foundations

- **Reinforcement Learning in Traffic Control:** *Noaeen et al. (2022)*, "Reinforcement Learning in Urban Network Traffic Signal Control: A Systematic Review."
- **Safe RL Mechanisms:** *SafeLight (AAAI 2023)*, "Safe Reinforcement Learning for Traffic Signal Control via Shielding Mechanisms."
- **Emergency Vehicle Preemption:** *Bieker-Walz & Behrisch (2019)*, "Green Waves for Emergency Vehicles in Urban Simulation."
- **Mixed-Traffic Computer Vision:** *India Driving Dataset (IDD)*, IIIT Hyderabad & Intel (46,500+ Indian road scenes).

Industry Standards & Protocols

- **SAE J2735 Standard:** Dedicated Short Range Communications (DSRC) / C-V2X Message Set Dictionary (SPaT / MAP).
- **USDOT CARMA-Streets Project:** Signal Phase and Timing JSON Message Specifications for Infrastructure-to-Vehicle (I2V).
- **German Aerospace Center (DLR):** SUMO Green Light Optimal Speed Advisory (GLOSA) Micro-Simulation Model.
- **Ultralytics YOLO11 + ByteTrack:** Real-Time SOTA Object Detection and Multi-Object Tracking.

Project Summary & Future Deployment Pathway

CoMIT proves that AI-driven traffic optimization can be deployed safely on heterogeneous Indian roads. By combining **Adaptive PPO Reinforcement Learning** with a **Deterministic Mathematical Safety Guardrail**, **Cryptographic Emergency Preemption**, and **Inclusive V2X Telemetry**, CoMIT delivers a 21% reduction in delays and 52% lower emissions while guaranteeing zero conflicting green hazards.

Next Phase: Deployment of the validated perception and control nodes onto physical Raspberry Pi 5 edge devices, connected roadside CSI cameras, and physical ESP32 in-cabin voice modules.